Emulating Rainfall–Runoff-Inundation Model Using Deep Neural Network with Dimensionality Reduction

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ABSTRACT: Predicting the spatial distribution of maximum inundation depth (depth-MAP) is important for the mitigation of hydrological disasters induced by extreme precipitation. However, physics-based rainfall–runoff-inundation (RRI) models, which are used operationally to predict hydrological disasters in Japan, require massive computational resources for numerical simulations. Here, we aimed at developing a computationally inexpensive deep learning model (Rain2Depth) that emulates an RRI model. Our study focused on the Omono River (Akita Prefecture, Japan) and predicted the depth-MAP from spatial and temporal rainfall data for individual events. Rain2Depth was developed based on a convolutional neural network (CNN) and predicts depth-MAP from 7-day successive hourly rainfall at 13 rain gauge stations in the basin. For training the Rain2Depth, we simulated the depth-MAP by the RRI model forced by 50 ensembles of 30-yr data from large-ensemble weather/climate predictions. Instead of using the input and output data directly, we extracted important features from input and output data with two dimensionality reduction techniques [principal component analysis (PCA) and the CNN approach] prior to training the network. This dimensionality reduction aimed to avoid overfitting caused by insufficient training data. The nonlinear CNN approach was superior to the linear PCA for extracting features. Finally, the Rain2Depth architecture was built by connecting the extracted features between input and output data through a neural network. Rain2Depth-based predictions were more accurate than predictions from our previous model (K20), which used ensemble learning of multiple regularized regressions for a specific station. Whereas the K20 can predict maximum inundation depth only at stations, our study achieved depth-MAP prediction by training only the single model Rain2Depth.

KEYWORDS: Flood events; Runoff; Deep learning; Dimensionality reduction

1. Introduction

Recent advances in high-performance computers (HPCs) have enabled numerical weather prediction (NWP) at high spatial resolution and with ensembles of >1000 members. For example, Yashiro et al. (2020) successfully conducted a global atmospheric data assimilation experiment at 3.5-km resolution with 1024 ensembles using the Fugaku, the flagship supercomputer of Japan. In the past decade, the computational resources of HPCs have been increased by adding cores of the central processing unit. For such many-core HPCs, increasing the number of ensemble members of NWP models is more scalable than reducing the horizontal and vertical resolutions because ensemble forecasts can be essentially parallelized. Therefore, a Japanese project is intended to prevent and mitigate weather-related disasters through the effective use of large-ensemble weather predictions.

The ensemble NWP forecasts enable probabilistic flood forecasts in which ensemble NWP forecasts are used as input for ensemble hydrological simulations. In the past two decades, extensive hydrological studies have been advancing knowledge of the physical processes of river discharge, runoff, and two-dimensional inundation (e.g., Sayama et al. 2012; Yamazaki et al. 2011). Recent studies carry out probabilistic flood forecasts using ensemble NWP data (e.g., Kobayashi et al. 2020). Owing to the progress of physical-based models, realistic simulation can be conducted for an operational flood warning system. However, the expanded physical-based models require progressively more and more computational resources, as observed in NWP models. For real-time flood warning systems, the exploration of computationally inexpensive inundation prediction methods is an important alternative to enhanced ensemble NWPs. For that purpose, this study uses deep neural network techniques.

Recently, applications of deep learning in NWP have been massively investigated. One application is to use deep learning for data-driven weather prediction such as precipitation nowcasting (e.g., Shi et al. 2017; Ravuri et al. 2021). Also, using deep learning in the postprocessing step to adjust model outputs is also known to be beneficial (e.g., Grönquist et al. 2021;

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Hess and Boers 2022). Combining deep learning and data assimilation is also well-suited to applications such as learning data assimilation (e.g., Bocquet et al. 2021; Tsuyuki and Tamura 2022) and learning observation operators (e.g., Liang et al. 2021). Challenging studies aim at emulating NWP models for learning relations between input and output weather data (Pathak et al. 2022; Keisler 2022). Namely, these studies use deep learning for emulating meteorological predictions, model output adjustments, data assimilation, observation operator, and atmospheric dynamical processes.

The emulation of physical-based models using neural networks has been also investigated broadly in the Earth sciences. For example, in atmospheric radiation studies, progress toward addressing the lack of computational resources has been made by moving from physical-based to machine learning–based models. Takenaka et al. (2011) and Shi et al. (2020) reconstructed the output (i.e., sky radiance and radiative flux) of the radiative transfer model; quasi-real-time processing of satellite observations was successfully achieved (e.g., Hashimoto and Nakajima 2017). Inspired by these previous studies, we aim to develop a deep learning–based inundation prediction method, called Rain2Depth, that emulates the calibrated physical-based inundation model. In the previous studies, most of the emulators of the inundation model have been developed for emulating the time series of the inundation depth using rainfall successively and can be classified into the following two types: emulating at a particular site (e.g., Mosavi et al. 2018) and in the area spatially (e.g., Chang et al. 2010; Lin et al. 2013; Jhong et al. 2017). The study on the emulator of a particular site has been investigated through more than 100 papers using various machine learning techniques in the last 2 decades (Mosavi et al. 2018).

In particular, we aimed to develop a deep learning–based surrogate model of the state-of-the-art inundation model, especially for evacuation plans in the early stages of rainfall. As compared with the abovementioned relevant studies, this study focuses on the maximum inundation depth during the event. Kotsuki et al. (2020) proposed a regression-based emulator of an inundation model and demonstrated good agreements with model-based predictions. However, the machine proposed by Kotsuki et al. (2020) can only predict inundation depth at stations. Therefore, we proposed a new machine that can predict the spatial pattern of inundation depth using a time series of distributed rainfall data. General deep neural networks, including networks in the aforementioned studies (Takenaka et al. 2011; Shi et al. 2020), require very large amounts of training data to optimize massive parameters in the network. In general, the high-resolution inundation simulation requires high-dimensional input data and produces high dimensional output data. However, for hydrological emulators, there would be fewer essential features within input rainfall data and output inundation patterns. Here, we propose training the network with features extracted by linear principal component analysis and the nonlinear neural network–based autoencoder (section 3a). We then build the Rain2Depth model by connecting input and output features, as described in section 3b. In this study, we apply the proposed machine to the Omono River, one of the most frequently flooded rivers in Japan; we compare the result with the physical-based model simulation.

This study is organized as follows. Section 2 describes the methods and experimental settings. Section 3 reveals the experimental results and provides discussion. Section 4 provides a summary.

2. Methods/experimental design

This study proposes a machine learning model, called Rain2Depth, that computes the spatial distributions of maximum inundation depth from input rainfall data (Fig. 1). Training inundation data for Rain2Depth were generated by a physical-based rainfall–runoff-inundation (RRI) model (Sayama et al. 2012), an operational real-time flood prediction model used in Japan (e.g., in Hyogo Prefecture). The Policy Decision Making for Future Climate Change (d4PDF; Mizuta et al. 2017) database was used for the input rainfall data, as described in section 2a. The Rain2Depth model is based on a neural network (section 2c). To enhance generality, we applied dimensionality reduction techniques to input and output data prior to neural network training (section 2b). The models were evaluated with cross validation, as described in section 2d.

a. Training data

1) Target river basin: Omono River basin

The Omono River, with a basin area of 4710 km^2, is a class-A river located in the northeastern part of Japan. It is one of the most frequently flooded rivers in Japan; it has been flooded four times since 2000. Therefore, the development of rapid flood forecasting systems is critical in this region. Historical floods have mainly been caused by the stationary baiu front.

2) Rainfall data of d4PDF with bias correction

Rainfall data obtained from d4PDF (Mizuta et al. 2017) were used as input for RRI simulations. The d4PDF provides the results of ensemble experiments that comprise more than 1000 years of meteorological data for both historical reproductions and future projections. Specifically, the historical experiments reproduced the past period of 1951–2010 with 20-km-resolution 50 ensembles, which was dynamically downscaled by 60-km global ensemble experiment driven by perturbed boundary conditions to sea surface and sea ice temperature. This study used these 50 ensemble experiments, with a focus on the past 30 years (1981–2010). We identified the event with the maximum 30-h precipitation in each year. For the RRI simulations, we extracted successive 168-h (i.e., 7 day) precipitation data including the heaviest precipitation: 48-h spin up, 30-h heavy precipitation, and 90-h rest periods. In total, 1500 events were extracted for RRI simulation.

Prior to the RRI simulation, a bias correction method (Watanabe et al. 2020) was applied to the d4PDF rainfall data for reducing unignorable bias in model-based precipitation data. The observation dataset of the Automated Meteorological Data Acquisition System (AMeDAS) operated by the Japan Meteorological Agency was used as reference data for the bias correction. This bias correction led to improved extreme

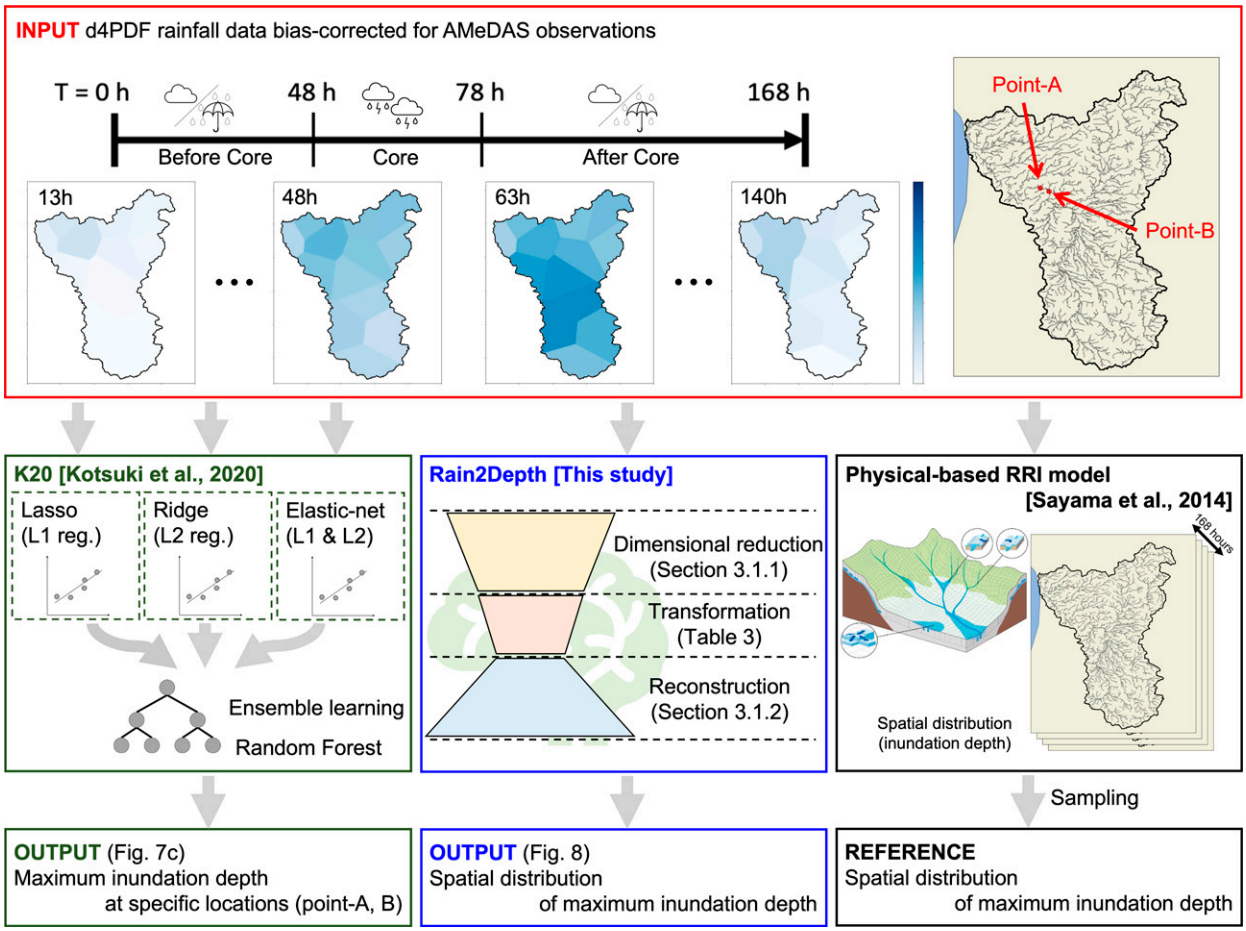

FIG. 1. Conceptual design of this study; input data were the rainfall data generated from d4PDF with the bias correction method (Watanabe et al. 2020) described in section 2a(2); reference output data were the spatial distribution of maximum inundation depth generated from the rainfall data with the physical-based RRI model (Sayama et al. 2012) described in section 2a(3); K20 and Rain2Depth are the emulators developed by Kotsuki et al. (2020) and this study, respectively. Points A and B in the figure are the target sites for Kotsuki et al. (2020).

precipitation events in the past. The operational design rainfall of the Omono River, which is 258.7 mm per 2-day period, was reproduced with an error of <10% using this bias correction for the d4PDF. In this procedure, the rainfall data were discretized spatially into 13 AMeDAS stational data of 168 h.

3) RRI MODEL AND MAXIMUM INUNDATION DEPTH

The RRI model is a two-dimensional physical-based distributed hydrological model, that simultaneously simulates both the rainfall–runoff process in slope areas and the flood inundation process in rivers and floodplains. In this section, we first describe the RRI model setting followed by experimental setting to produce spatial inundation depth training data.

This study used the RRI model calibrated and validated by Abe et al. (2019) in the Omono River. The spatial resolution of the model is 270 m. All grid cells with an upstream area of >2 km^2 were regarded as river grid cells. For the river geometry of main rivers, we included the cross-section shapes of river channels surveyed by the Ministry of Land, Infrastructure, Transportation and Tourism of Japan. In addition, we modeled the flood control operations of six dams in the Omono River (Konja et al. 2018). To model runoff processes, we applied the unsaturated lateral flow mechanism for forest area and the saturated lateral flow mechanism for other land-use areas. The model was calibrated with two heavy rainfall events that occurred in 2004 and 2011; it was validated by focusing on the record-breaking flood in 2017. The RRI model showed good reproducibility in terms of the observed discharge, inundation area and depth (Abe et al. 2019). For all six official discharge observation points, Nash–Sutcliff efficiency values were >0.70. The reproducibility of inundated and noninundated areas was 90%. See Abe et al. (2019) for more details.

Using the calibrated RRI model, we conducted experiments with the bias-corrected d4PDF rainfall data for 1500 events. We used the maximum inundation depth of each grid to generate the spatial distribution of maximum inundation depth (depth-MAP) for each event.

TABLE 1. Architecture of the convolutional autoencoder for rainfall data. Here and in subsequent tables, the numbers in brackets indicate the dimensions of the array in the program.

Layer type	Output channel	Kernel size	Padding	Stride
Encoder (Input shape: [13, 1, 168])				
Conv1d + ReLU	[13, 16, 168]	3	1	1
Conv1d + ReLU	[13, 16, 168]	3	1	1
Conv1d + ReLU	[13, 16, 168]	3	1	1
Conv1d + ReLU	[13, 16, 168]	3	1	1
Conv1d + ReLU	[13, 16, 84]	2	0	2
Conv1d + ReLU	[13, 16, 84]	3	1	1
Conv1d + ReLU	[13, 16, 84]	3	1	1
Conv1d + ReLU	[13, 16, 28]	3	0	3
Conv1d + ReLU	[13, 16, 14]	2	0	2
Linear + LayerNorm	[13, 81]	—	—	—
Decoder (Input shape: [13, 81])				
Linear	[13, 224]	—	—	—
ConvTranspose1d + ReLU	[13, 16, 28]	2	0	2
ConvTranspose1d + ReLU	[13, 16, 84]	3	0	3
Conv1d + ReLU	[13, 16, 84]	3	1	1
Conv1d + ReLU	[13, 16, 84]	3	1	1
ConvTranspose1d + ReLU	[13, 16, 168]	2	0	2
Conv1d + ReLU	[13, 16, 168]	3	1	1
Conv1d + ReLU	[13, 16, 168]	3	1	1
Conv1d + ReLU	[13, 16, 168]	3	1	1
Conv1d	[13, 16, 168]	3	1	1

b. *Dimensionality reduction techniques*

Deep neural network training processes require massive training data because such processes involve large numbers of network parameters depending on the features of the input and output data. However, this study endeavored to train the network with a moderately small number of events (1500) as training data. Therefore, it was beneficial to extract important features from input (rainfall) and output (depth MAP) data, then use these extracted features to train a network with reduced network parameters. Here, we extracted important features by reducing the data dimensionality.

The easiest way to reduce data dimensionality is principal component analysis (PCA); this method reduces the dataset features after orthogonalization by singular value decomposition. PCA is known as statistical recognition tools as empirical orthogonal functions (Wu et al. 2009) or proper orthogonal decomposition (Lumley 1967) in meteorology and geophysical fluid dynamics fields, for example, postprocessing tools (Murray and Ukeiley 2007) and dominant components analysis (Kikuchi et al. 2016). However, PCA would be suboptimal choice if major modes are nonlinear. An alternative method is a neural network–based technique known as autoencoder (AE), which can extract features through nonlinear multilayer neural networks with activation functions. In this study, we used the convolutional–neural network (CNN) AE (CNN-AE). It uses CNN networks as encoder and decoder before and after feature extraction using a fully connected layer. Because the networks (e.g., activation function, normalization, and layer numbers) were built empirically using training and validation (TRAIN/VAL) data, there might still be room for optimization. This study compared the efficiency of data reduction between PCA and AE methods.

The present study designed the AE as follows. The CNN-AE for input rainfall data was constructed by one-dimensional convolution (Conv1d). Conv1d was developed for electrocardiogram classification (Kiranyaz et al. 2015). Recent studies have used Conv1d for time series data in geophysics (e.g., Makinoshima et al. 2021; Van et al. 2020) because of its low computational cost (Kiranyaz et al. 2021). For rainfall–runoff emulation, Conv1d may be more suitable than long short-term memory (Hochreiter and Schmidhuber 1997) because Conv1d effectively extracts the dependencies (features) in short-term time series (Van et al. 2020). Therefore, the features of the rainfall data were extracted using the Conv1d, fully connected layer ("Linear" in the tables), rectified linear units (ReLU; Nair and Hinton 2010), and layer normalization ("LayerNorm" in the tables; Ba et al. 2016), as shown in Table 1. The CNN-AE for the rainfall data was constructed using the one-dimensional convolutional layers and a fully connected layer in the encoder, while one-dimensional transposed convolutional layers and a fully connected layer were used in the decoder. Layer normalization can constrain the network parameters by specific layer data without batch data (Ba et al. 2016). Therefore, it is applicable for data with both large and small batch sizes. Layer normalization is also suitable for large variance data, such as the extreme weather data used in this study; other normalization techniques (e.g., batch normalization; Ioffe and Christian 2015) are not suitable for large variance data.

Two-dimensional convolutional neural networks for image data are rapidly progressing technologies (e.g., Krizhevsky et al. 2012). A two-dimensional convolutional layer (Conv2d) convolves the neighbor pixels of a target pixel and, thus, potentially extracts spatially distributed local features. These characteristics would be beneficial for the feature extraction of depth-MAP

TABLE 2. Architecture of the convolutional autoencoder for depth-MAP.

Layer type	Output channel	Kernel size	Padding	Stride
Encoder (Input shape: [1, 1, 432, 324])				
Conv2d + ReLU	[1, 128, 216, 162]	2	0	2
Conv2d + ReLU	[1, 128, 108, 81]	2	0	2
Conv2d + ReLU	[1, 128, 36, 27]	3	0	3
Conv2d + ReLU	[1, 128, 12, 9]	3	0	3
Conv2d + ReLU	[1, 128, 4, 3]	3	0	3
Linear + LayerNorm	[1, 10]	—	—	—
Decoder (Input shape: [1, 10])				
Linear	[1, 1536]	—	—	—
ConvTranspose2d + ReLU	[1, 128, 12, 9]	3	0	3
ConvTranspose2d + ReLU	[1, 128, 36, 27]	3	0	3
ConvTranspose2d + ReLU	[1, 128, 108, 81]	3	0	3
ConvTranspose2d + ReLU	[1, 128, 216, 162]	2	0	2
ConvTranspose2d	[1, 1, 432, 324]	2	0	2

because the inundated area should be continuously distributed around the river. In this study, we constructed the CNN-AE for depth-MAP data by using a two-dimensional convolutional neural network (Table 2).

c. *Neural network architectures*

To emulate the maximum inundation depth from rainfall data (i.e., Rain2Depth), we constructed a neural network by connecting the features of rainfall and the depth-MAP data extracted by dimensionality reduction techniques, as described in section 2b. The network consisted of the one-dimensional convolutional network, ReLU, and the fully connected layer, as shown in Table 3.

d. *Cross validation*

The dataset was grouped into 1000 data for training (labeled as TRAIN)/validation (labeled as VAL) and 500 data for tests (labeled as TEST), as shown in Fig. 2. TEST data were not used for emulator validation and architecting, as detailed in sections 3a and 3b(1). After the network had been architected [section 3b(2)], the pseudo generalization performance of the network was estimated from TEST data that consist of independent data from the TRAIN/VAL training procedure. Validations of the models were performed via fivefold cross validation (5FCV) using TRAIN/VAL and a small training dataset that allowed assessment of training model feasibility. The 5FCV divides the TRAIN/VAL into five subsets; it uses four subsets (total of 800 data) for TRAIN and the remaining subset (200 data) for VAL (Fig. 2a). Therefore, the evaluation score of the network was determined from the mean of five trials. For the validation of the small set of training data, the score of the network was determined from the evaluation score of each trial derived from subtrials (Fig. 2b). For example, when 400 data were used for training, the score of each trial was derived from 6 [i.e., combination C(4, 2)] subtrials.

e. *Brief outline of an emulator with multiple regularized regressors*

Kotsuki et al. (2020) emulated the maximum inundation depth at the two locations denoted as points A and B in Fig. 1 through ensemble learning with multiple regularized regressors (herein this method is referred to as K20). The architecture of their emulator is illustrated in Fig. 1. It consists of the following two aspects: maximum inundation depth at the specific location predicted by three regressors with different regularization [ridge, least absolute shrinkage and selection operator (LASSO), and elastic net] and ensemble learning using a random-forest classifier. In this study, K20 was trained using TRAIN/VAL data (1000 data) as described by Kotsuki et al. (2020) with the hyperparameters of regularization determined in their paper.

3. Results and discussion

a. *Dimensionality reduction*

1) RAINFALL DATA

Because the rainfall data of 13 AMeDAS stations contained hourly data over seven days, the rainfall data had two

TABLE 3. Architecture of the convolutional neural network of the middle layer in Rain2Depth.

Layer type	Output channel	Kernel size	Padding	Stride
(Input shape: [1, 81, 13])				
Conv1d + ReLU	[1, 200, 13]	1	0	1
Conv1d + ReLU	[1, 200, 13]	1	0	1
Conv1d + ReLU	[1, 200, 13]	1	0	1
Conv1d + ReLU	[1, 200, 13]	1	0	1
Conv1d + ReLU	[1, 20, 1]	13	0	1
Linear	[1, 10]	—	—	—

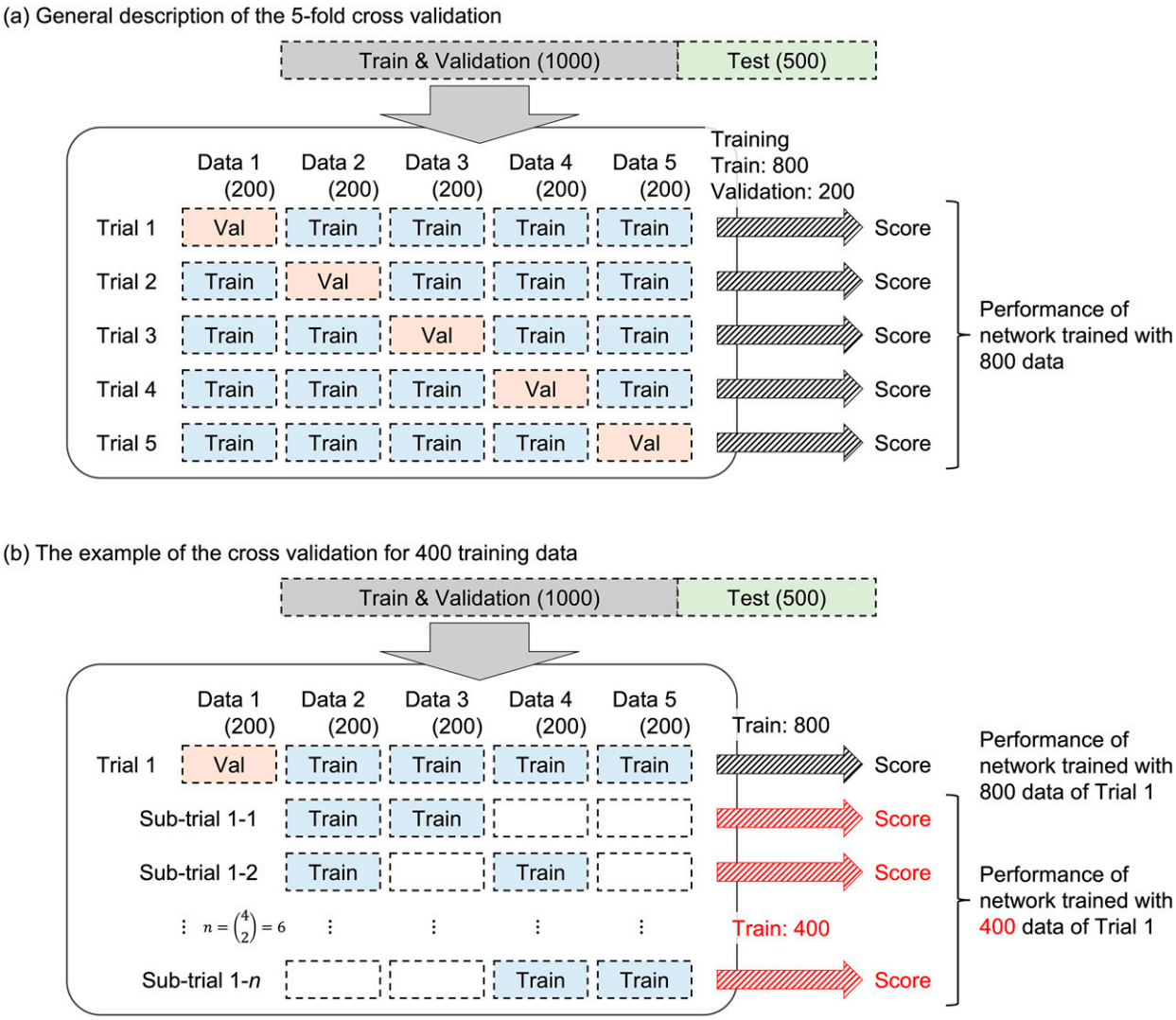

FIG. 2. Concept of fivefold cross validation: (a) general description of 5FCV and (b) description for small numbers of data points.

dimensions and 2184 variables (168 h × 13 AMeDAS stations) in each event (herein, rain-2D data). This number exceeded the number of training data; thus, we reduced dimensionality by using PCA. The dashed lines in Fig. 3 show the PCA results for 200–800 training data. The root-mean-squared errors (RMSEs) of TRAIN and VAL decreased as the number of training data increased; the use of 800 data for training yielded the best performance. An increased number of principal components was expected to improve the performance of TRAIN rainfall data reconstruction, but the TRAIN rainfall data did not match the VAL rainfall data because of overfitting. When 200 training data were used, the VAL performance was not significantly improved with respect to TRAIN performance when the number of principal components was >10. These findings suggest that principal components beyond the 10th component do not enhance generalization performance. These findings were also observed for principal components beyond the 20th component when 800 training data were used, suggesting that the dimension of the rain-2D data is significantly increased beyond 800 data. Therefore, additional training data for dimensionality reduction are needed to retain generalization ability. This feature arises from two factors: the temporal distribution of rainfall (i.e., time series of rainfall) and spatial distribution of rainfall of the 13 AMeDAS stations (e.g., change in rainfall region position from south to north or another movement). If the features of the temporal distribution of rainfall are identical at all AMeDAS stations, the dimension of the rainfall data observed at each AMeDAS station (herein, rain1d) can be reduced with the same machine. This assumption also produces 13-fold more effective training data relative to rain-2D data. The dimension of the rain-1D data can be reduced while retaining generalization performance even when using 200 training data (solid lines in Fig. 3). Thus, the features of the temporal distribution of the rainfall (such as rain and stop mechanism) are similar across all stations.

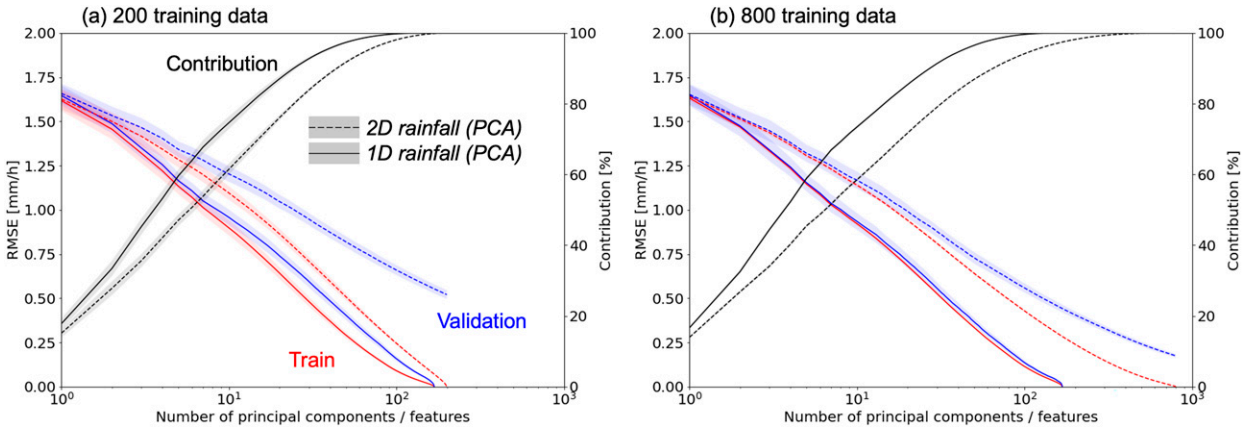

FIG. 3. Relationship between the number of principal components and the RMSE for rainfall data. Solid and dotted lines are the results from rain-1D and rain-2D data, respectively. Black lines indicate the cumulative contribution ratio.

As compared with PCA, the neural network approach can efficiently reduce the number of feature values through the use of an activation function (ReLU) and convolutional layers. As discussed above, we only conducted dimensionality reduction of the rain-1D data with CNN-AE. Figure 4 shows the results of the PCA and CNN-AE approaches for the rain-1D data. The RMSE values of VAL data obtained by the CNN-AE were lower than the values obtained by PCA for the rain-1D data when the number of features was >20. In other words, the nonlinear NN-based AE would not always outperform the PCA when the number of features is too small. Therefore, this result indicates the importance of comparing NN-based AE with PCA for identifying whether NN-based AE is more suitable than PCA. Architecture optimization for a low number of features may enable the CNN-AE to extract more features than PCA when there is a low number of features (e.g., 10), but investigating this point is beyond the scope of present study.

2) SPATIAL DISTRIBUTION OF MAXIMUM INUNDATION DEPTH

In the previous section, the dimension of the rainfall data was reduced by solely focusing on the features of the temporal distribution of rainfall data at individual AMeDAS stations (rain-1D data). However, the feature extraction of the depth-MAP is likely to be overfitted because it requires direct reduction of the dimension of the depth-MAP. Furthermore, there is no method for increasing the amount of effective training data other than by increasing the amount of ensemble simulation data, although the number of variables is significantly more in the depth-MAP than in the rainfall data. Figure 5 shows the results of PCA and CNN-AE for the depth-MAP. Relative to the cumulative contribution of PCA for the rain-2D data, the contribution of PCA for the depth-MAP was high for an identical number of principal components. When 800 training data were used, 112 and 5 features were necessary to extract the principal components of the

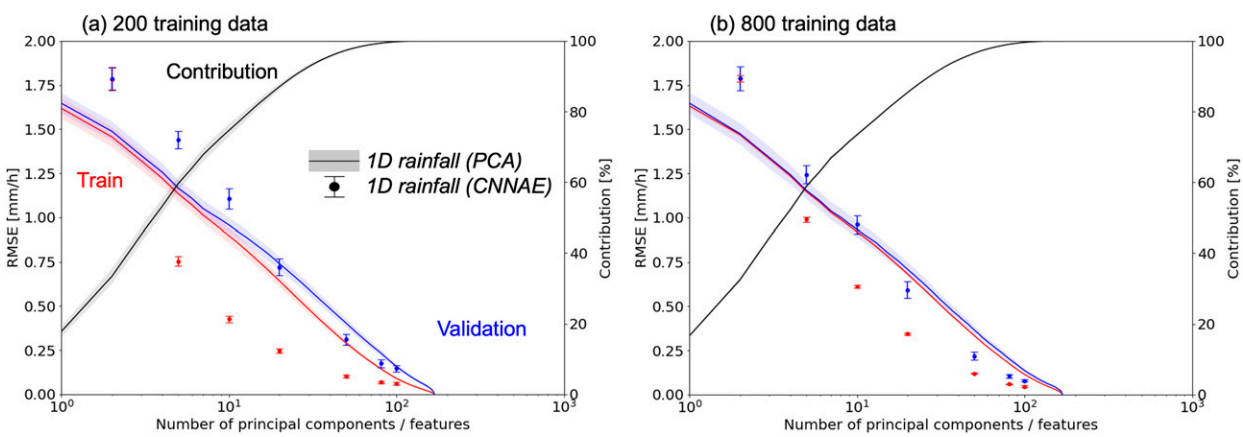

FIG. 4. Relationship between the number of principal components (features) and the RMSE values derived for CNN-AE and PCA for rainfall data. Solid lines and dots are the results derived by PCA and CNN-AE for rain-1D data, respectively. Black lines indicate the cumulative contribution ratio for PCA.

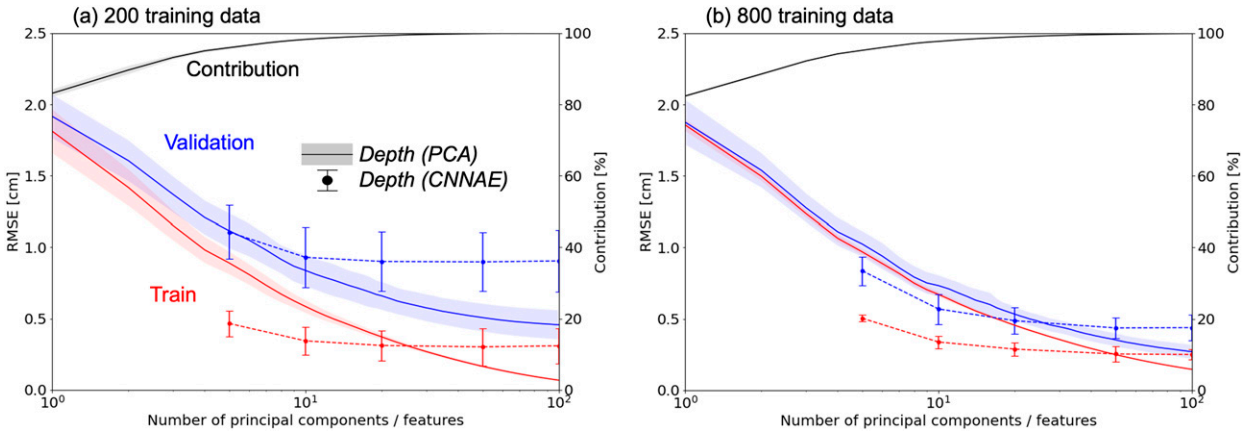

FIG. 5. As in Fig. 4, but for depth-MAP data.

rain-2D and depth-MAP data within a 95% cumulative contribution, respectively. This finding suggests that the number of important features in the depth-MAP is significantly lower; moreover, the depth-MAP obtained from the physical-based RRI model was nearly reconstructed with a minimum of 5 features. The RMSE of the PCA for the depth-MAP decreased as the number of training data increased, similar to the results of PCA for the rain-1D data. The CNN-AE for the depth-MAP consists of the two-dimensional convolutional layers (before feature extraction by the fully connected layer in the encoder), as well as the two-dimensional transposed convolutional layers in the decoder (Table 2). Although there remains challenges in optimizing the CNN-AE architecture, the CNN-AE extracts features more effectively than does PCA when the number of neurons in the fully connected layer is <20 with 800 training data.

b. *Emulation of the physical-based RRI model with the neural network Rain2Depth*

Rain2Depth is an emulator of a physical-based RRI model and consists of three parts: the feature extraction layer for rain-1D data, the feature connecting layer from the rain-1D data to depth-MAP, and the reconstructing layer for the depth-MAP (Table 3). The numbers of features of the rain-1D and Depth-MAP were empirically determined to be 10 and 81 form the RMSE decay curve (Figs. 4 and 5). Section 3b(1) evaluates Rain2Depth with 5FCV through comparison between the PCA and CNN-AE during feature extraction for the rain-1D and depth-MAP. Then, section 3b(2) provides the result of the application of the TEST data and compares the maximum inundation depth at point A with K20.

1) EVALUATION WITH 5FCV

This section evaluates Rain2Depth by comparing the dimensionality reduction techniques of PCA and CNN-AE. Figure 6 shows the RMSE of the depth-MAP predicted by Rain2Depth with PCA and CNN-AE. The RMSE of VAL predicted by Rain2Depth was larger than the RMSE predicted by PCA and CNN-AE, as demonstrated in Fig. 5. This is attributable to the nonlinear relationship of features between the rain-1D data and the depth-MAP data. By increasing the number of training data, the RMSE improved from 2.92 (200 data) to 2.18 cm (800 data)

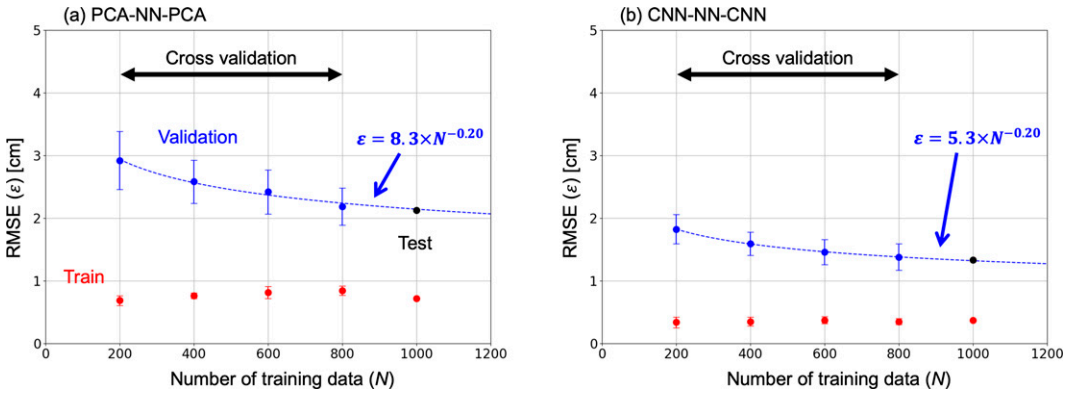

FIG. 6. Relationship between training data size and RMSE of depth-MAP emulated with dimensional reduction using (a) PCA and (b) CNN-AE. Red, blue, and black dots show the results from TRAIN, VAL, and TEST data, respectively. Error bars show the standard deviation of 5FCV; The blue dotted line shows a power-law fit line from the RMSE of VAL data.

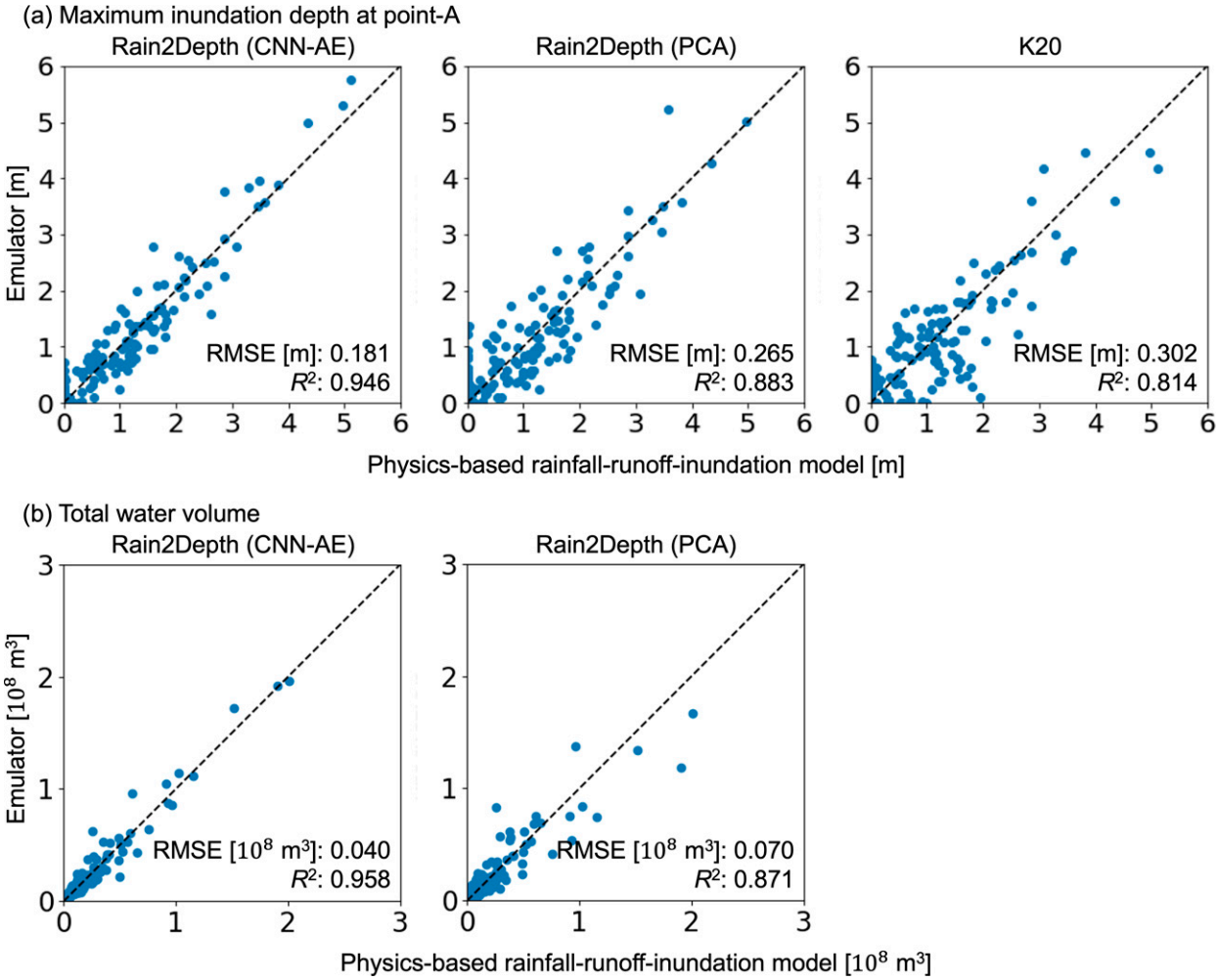

FIG. 7. Example of TEST data for (a) the maximum inundation depth at point-A and (b) the total water volume in the target area (red-outlined box in Fig. 8a, below), obtained using Rain2Depth with CNN-AE, Rain2Depth with PCA, and K20.

in Rain2Depth with PCA; it improved from 1.82 (200 data) to 1.38 cm (800 data) in Rain2Depth with CNN-AE. The larger residual errors related to nonlinearity could be reduced by using a very large amount of data to train Rain2Depth. Comparison of dimensionality reduction techniques showed that the RMSE of the CNN-AE was smaller than the RMSE of the PCA because the CNN-AE technique can extract more information than the PCA technique with the same number of features. When the RMSE meets the power law, using a training data number N of $8.3 \times N^{-0.2}$ for PCA and $5.3 \times N^{-0.2}$ for CNN-AE, $\sim 4 \times 10^4$ training data for Rain2Depth with PCA and $\sim 4 \times 10^3$ training data for Rain2Depth with CNN-AE were required to attain an RMSE of 1 cm in VAL, similar to the RMSE of TRAIN. For an RMSE of 1.4 cm in VAL, which is identical to the RMSE for Rain2Depth with CNN-AE using 800 training data, the Rain2Depth with PCA required $\sim 7.5 \times 10^3$ training data. Therefore, our approach (i.e., Rain2Depth with CNN-AE) can train a network more efficiently, relative to the other emulators described in this paper.

2) PERFORMANCE WITH TEST DATA

This study aimed to reduce computational costs by emulating the physical-based RRI model. Through 5FCV for several data sizes, we determined that a very large amount of training data is needed to predict within 1 cm accuracy the maximum inundation depth when >1000 data are used. In this section, we describe the performance of Rain2Depth trained by all TRAIN/VAL data (total of 1000 data) and apply it to the TEST data (500 data), which are independent from the Rain2Depth training data, as described in section 2d. We also conducted a comparison of the maximum inundation depth at point A with the result of a previous study (K20), as described in section 2e.

The RMSE for the 1000 training data is shown in Fig. 6. The RMSE of TEST using Rain2Depth with PCA and CNN-AE meets the power law fitted according to the results of 5FCV, which combines 1000 ensemble data for TRAIN and VAL, independent of TEST. This result indicates that the evaluation using VAL data is meaningful. Figure 7a shows the

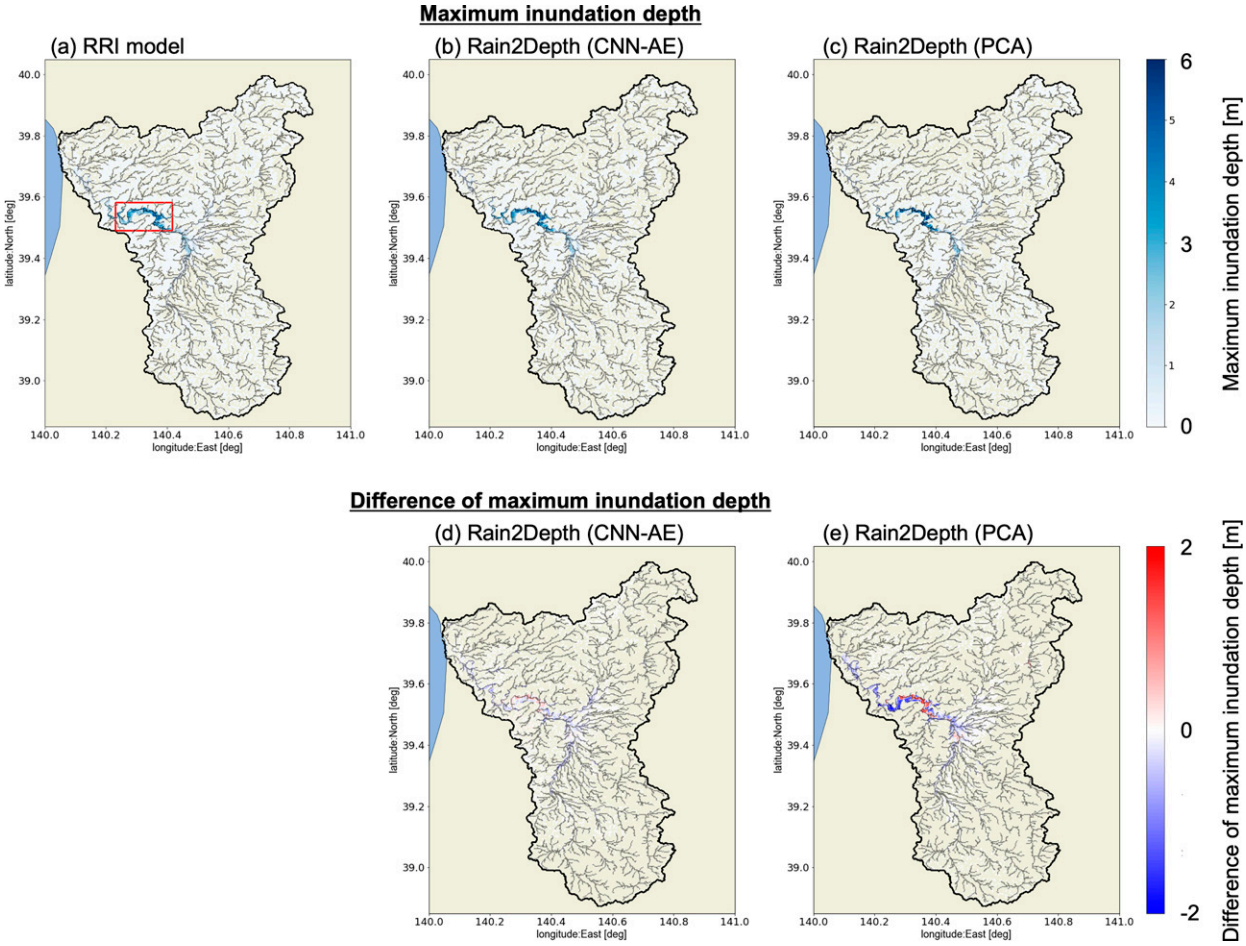

FIG. 8. Example of the depth-MAP obtained using (a) the physics-based RRI model, (b) Rain2Depth with CNN-AE, and (c) Rain2Depth with PCA. Also shown is the spatial distribution of the difference between the physics-based RRI model and (d) Rain2Depth with CNN-AE or (e) Rain2Depth with PCA. The red-outlined box is the target area, which has frequent inundation.

performance of the maximum inundation depth at point A emulated by Rain2Depth with PCA and CNN-AE, as well as K20, for TEST data. The RMSE of the maximum inundation depth at point-A is larger than the result in Fig. 6. This is because Fig. 6 includes the noninundated pixels in all the events, whereas point-A is an extremely inundated area. This indicates that the reconstructing in inundated areas has a difficulty by an emulator trained with the small data. The maximum inundation depth obtained using Rain2Depth is better than the result obtained using K20, possibly because of nonlinear transformation by the middle network in Rain2Depth (Table 3). For the total water volume in the target area, which has frequent inundation (Fig. 8), Rain2Depth with CNN-AE has the better agreement (Fig. 7b) with a physical-based model than PCA because CNN-AE trained the relationship with neighbor pixels by convolutional layers.

Figure 8 shows the example of the depth-MAP of TEST data obtained using Rain2Depth with PCA and CNN-AE. There is inundation around the Omono River, especially at serpentine and downstream locations (Fig. 8a). These inundation characteristics were retrieved by Rain2Depth with both PCA and CNN-AE. The result shows Rain2Depth with CNN-AE had a potential to predict the spatial distribution of the maximum inundation depth better than Rain2Depth with PCA.

4. Conclusions

In this study, we developed an emulator, Rain2Depth, of the physical-based RRI model with a deep convolutional neural network. This network consists of feature extraction of the rainfall data, feature transposition from the rainfall to the spatial distribution of the maximum inundation depth, and reconstruction of the maximum inundation depth using spatial distribution features. To extract the rainfall and inundation features, we used two approaches: PCA and CNN-AE. Because CNN-AE can extract nonlinear features, the RMSE of the reconstructed data with CNN-AE was smaller than the RMSE with PCA. Thus, the dimensionality reduction by CNN-AE was suitable for constructing an emulator of the physical-based RRI model while maintaining a small network trained with a small number of training data.

Rain2Depth was constructed using a deep neural network to connect the features of input and output variables. The results

according to 5FCV indicated that Rain2Depth with CNN-AE is better than Rain2Depth with PCA when using the same number of features. In comparison with the previous study (K20), in terms of maximum inundation depth at a specific location, we showed that Rain2Depth with CNN-AE was the best emulator among the three models because the middle layer in Rain2Depth provides nonlinear transformation. Thus, by using the middle network in Rain2Depth (instead of K20) to emulate the maximum inundation depth at point A, we achieved better performance than K20 but worse performance than Rain2Depth with CNN-AE. Unlike K20, Rain2Depth can predict the spatial distribution of maximum inundation depth.

For future work, we have the three goals: evaluation of the emulator through application to the actual inundation event (e.g., July 2017), optimization of the network with a small number of training data, and application to more complicated physics processes (e.g., time series of the spatial distribution of inundation depth). Such improvements may enable the establishment of an operational flood warning system.

Japan suffers from an increasing trend of floods and inundations induced by severe precipitation. Improving real-time inundation predictions is important to mitigate such disasters from reservoir operations and provide sufficient time for residents' evacuation. For that purpose, operational real-time systems have been installed for major rivers in Japan using physical rainfall–runoff-inundation models with assimilation of water depth observation. There can be two possible applications to bring our methods to operational systems.

The first application is the use for computationally inexpensive inundation predictions. The second application is the use for the data assimilation system of the RRI models. Because of the nonlinearity of rainfall–runoff, river, and inundation processes, particle filters would be a more suitable data assimilation method for RRI models than ensemble Kalman filters and variational methods. However, the particle filter requires many more ensembles with more computations for employing RRI simulations. Here emulators could be used for increasing ensembles as augmented members (Kotsuki and Bishop 2022). In addition to such methodological developments, tight discussions with operational administrations are also necessary. Such further updates toward operational systems are important directions of our future studies.

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Data availability statement. Because of the large volume of involved data and limited disk space available, the data will be shared online upon request to author Masahiro Momoi (momoi-masahiro@doerresearch.com).

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